

T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1 - S_2 contraction
$\lambda_{S_1}^{(A)}$	additional independent S_1 self-contraction
$\lambda_{S_2}^{(A)}$	additional independent S_2 self-contraction
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$
$T(d)$	$d(d^2 - 1)/12$
$C_{LLS_1 S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$
Quantity: g_1

$$16\pi^2\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$$

Column Representative term structure	
T1	$\frac{43g_1^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_1^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_1^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{59g_1^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{59g_1^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{59g_1^3}{6}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{17g_2^3}{6}$
T2	$-\frac{g_2^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17g_2^3}{6}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{7g_2^3}{6}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-\frac{g_2^3}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{g_2^3}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\frac{g_2^3}{6}$

$\beta_{g_3}^{(1)}$
Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column		Representative term structure
T1		$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$-3\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$
T2	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure	
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	
T3	$-8g_3^2\ (Y_u)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

Quantity: Y_d

$16\pi^2\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$

Quantity: Y_e

$$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$
T3	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$

Quantity: Y_e

Terms 3 of 3

Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{y_1}^{(1)}$
Quantity: y_1

$$16\pi^2\,\mu\,\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)\,(y_2)_{ij}$
T2	$+T_\nu^{(1)}\,(y_1)_{ij}$
T3	$+\frac{1}{2}(2\,(y_1\ (y_2)^\dagger\ y_2)_{ij}+(y_2\ (y_2)^\dagger\ y_1)_{ij}+(Y_e\ (Y_e)^\dagger\ y_1)_{ij}+3\,(y_1\ (y_1)^\dagger\ y_1)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+T_\nu^{(1)}\,(y_1)_{ij}$	$+\frac{1}{2}(2\,(y_1\ (y_2)^\dagger\ y_2)_{ij}+(y_2\ (y_2)^\dagger\ y_1)_{ij}+(Y_e\ (Y_e)^\dagger\ y_1)_{ij}+3\,(y_1\ (y_1)^\dagger\ y_1)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$T_\nu^{(1)}\,(y_1)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$T_\nu^{(1)}\,(y_1)_{ij}$	—

Quantity: y_1

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9}{4}g_2^2(y_1)_{ij}$
T2	$-\frac{3}{4}g_1^2(y_1)_{ij}$
T3	$\frac{3}{2}\text{Tr}\left(y_1(y_2)^\dagger\right)(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9}{4}g_2^2(y_1)_{ij}$	$-\frac{3}{4}g_1^2(y_1)_{ij}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{33}{4}g_2^2(y_1)_{ij}$	$-\frac{3}{4}g_1^2(y_1)_{ij}$	$\frac{3}{2}\text{Tr}\left(y_1(y_2)^\dagger\right)(y_2)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_1^2(y_1)_{ij}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_1^2(y_1)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_1^2(y_1)_{ij}$	—

Quantity: y_1

Terms 3 of 5

Column	Representative term structure
T1	$+ \frac{3}{2} T_{\nu}^{(1)} (y_1)_{ij}$
T2	$+ \frac{1}{4} (2 \left(y_1 \ (y_2)^{\dagger} \ y_2\right)_{ij} + 3 \left(y_2 \ (y_2)^{\dagger} \ y_1\right)_{ij} + 2 \left(Y_e \ (Y_e)^{\dagger} \ y_1\right)_{ij} + 5 \left(y_1 \ (y_1)^{\dagger} \ y_1\right)_{ij})$
T3	$- 2 T_{\nu}^{(1)} (y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3}{2} T_{\nu}^{(1)} (y_1)_{ij}$	$+ \frac{1}{4} (2 \left(y_1 \ (y_2)^{\dagger} \ y_2\right)_{ij} + 3 \left(y_2 \ (y_2)^{\dagger} \ y_1\right)_{ij} + 2 \left(Y_e \ (Y_e)^{\dagger} \ y_1\right)_{ij} + 5 \left(y_1 \ (y_1)^{\dagger} \ y_1\right)_{ij})$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$2 T_{\nu}^{(1)} (y_1)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+ \frac{3}{4} \left(y_2 \left(y_2 \right)^{\dagger} y_1 \right)_{ij}$
T2	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_1 \right)_{ij}$
T3	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3}{4} \left(y_2 \left(y_2 \right)^{\dagger} y_1 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_1 \right)_{ij}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: y_1

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{1}{2}((y_2\ (y_2)^\dagger\ y_1)_{ij} + (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 3\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$
T2	$+\frac{1}{4}(3\ (y_2\ (y_2)^\dagger\ y_1)_{ij} + 2\ (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 6\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{2}((y_2\ (y_2)^\dagger\ y_1)_{ij} + (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 3\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{1}{4}(3\ (y_2\ (y_2)^\dagger\ y_1)_{ij} + 2\ (Y_e\ (Y_e)^\dagger\ y_1)_{ij} + 6\ (y_1\ (y_1)^\dagger\ y_1)_{ij})$

$\beta_{y_2}^{(1)}$
Quantity: y_2

$$16\pi^2\,\mu\,\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\text{Tr}\big(y_2\,(y_1)^\dagger\big)\,(y_1)_{ij}$
T2	$+T_\nu^{(2)}\,(y_2)_{ij}$
T3	$+\frac{1}{2}((y_1\,(y_1)^\dagger\,y_2)_{ij}+2\,(y_2\,(y_1)^\dagger\,y_1)_{ij}+(Y_e\,(Y_e)^\dagger\,y_2)_{ij}+3\,(y_2\,(y_2)^\dagger\,y_2)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\big(y_2\,(y_1)^\dagger\big)\,(y_1)_{ij}$	$+T_\nu^{(2)}\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\,(y_1)^\dagger\,y_2)_{ij}+2\,(y_2\,(y_1)^\dagger\,y_1)_{ij}+(Y_e\,(Y_e)^\dagger\,y_2)_{ij}+3\,(y_2\,(y_2)^\dagger\,y_2)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$T_\nu^{(2)}\,(y_2)_{ij}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$T_\nu^{(2)}\,(y_2)_{ij}$	—

Quantity: y_2

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9}{4}g_2^2(y_2)_{ij}$
T2	$-\frac{3}{4}g_1^2(y_2)_{ij}$
T3	$\frac{3}{2}\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9}{4}g_2^2(y_2)_{ij}$	$-\frac{3}{4}g_1^2(y_2)_{ij}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{33}{4}g_2^2(y_2)_{ij}$	$-\frac{3}{4}g_1^2(y_2)_{ij}$	$\frac{3}{2}\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_2)_{ij}$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_2)_{ij}$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9}{2}g_2^2(y_2)_{ij}$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—

Terms 3 of 5

Column	Representative term structure
T1	$+ \frac{3}{2} T_{\nu}^{(2)} (y_2)_{ij}$
T2	$+ \frac{1}{4} (3 \left(y_1 \ (y_1)^{\dagger} \ y_2\right)_{ij} + 2 \left(y_2 \ (y_1)^{\dagger} \ y_1\right)_{ij} + 2 \left(Y_e \ (Y_e)^{\dagger} \ y_2\right)_{ij} + 5 \left(y_2 \ (y_2)^{\dagger} \ y_2\right)_{ij})$
T3	$+ \frac{1}{2} ((y_1 \ (y_1)^{\dagger} \ y_2)_{ij} + \left(Y_e \ (Y_e)^{\dagger} \ y_2\right)_{ij} + 3 \left(y_2 \ (y_2)^{\dagger} \ y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{3}{2} T_{\nu}^{(2)} (y_2)_{ij}$	$+ \frac{1}{4} (3 \left(y_1 \ (y_1)^{\dagger} \ y_2\right)_{ij} + 2 \left(y_2 \ (y_1)^{\dagger} \ y_1\right)_{ij} + 2 \left(Y_e \ (Y_e)^{\dagger} \ y_2\right)_{ij} + 5 \left(y_2 \ (y_2)^{\dagger} \ y_2\right)_{ij})$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+ \frac{1}{2} ((y_1 \ (y_1)^{\dagger} \ y_2)_{ij} + \left(Y_e \ (Y_e)^{\dagger} \ y_2\right)_{ij} + 3 \left(y_2 \ (y_2)^{\dagger} \ y_2\right)_{ij})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$2T_{\nu}^{(2)}(y_2)_{ij}$
T2	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$
T3	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$2T_{\nu}^{(2)}(y_2)_{ij}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: y_2

Terms 5 of 5

Column	Representative term structure
T1	$+ \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij}$
T2	$+ \frac{1}{4} (3 \left(y_1 \ (y_1)^\dagger \ y_2\right)_{ij} + 2 \left(Y_e \ (Y_e)^\dagger \ y_2\right)_{ij} + 6 \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+ \frac{1}{4} (3 \left(y_1 \ (y_1)^\dagger \ y_2\right)_{ij} + 2 \left(Y_e \ (Y_e)^\dagger \ y_2\right)_{ij} + 6 \left(y_2 \ (y_2)^\dagger \ y_2\right)_{ij})$

$\beta_{M_F}^{(1)}$

Quantity: M_F

$16\pi^2\,\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\left((M_F)^T\,(y_1)^\dagger\,y_1\right)_{ij}$
T2	$+ \left((y_1)^T\,(y_1)^*\,M_F\right)_{ij}$
T3	$+ \left((M_F)^T\,(y_2)^\dagger\,y_2\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\left((M_F)^T\,(y_1)^\dagger\,y_1\right)_{ij}$	$+ \left((y_1)^T\,(y_1)^*\,M_F\right)_{ij}$	$+ \left((M_F)^T\,(y_2)^\dagger\,y_2\right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: M_F

Terms 2 of 4

Column	Representative term structure
T1	$+ \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij}$
T2	$\frac{1}{2} \left(\left((M_F)^T \, (y_1)^\dagger \, y_1 \right)_{ij} + \left((y_1)^T \, (y_1)^* \, M_F \right)_{ij} + \left((M_F)^T \, (y_2)^\dagger \, y_2 \right)_{ij} + \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij} \right)$
T3	$- 6 g_2^2 \left(\left((M_F)^T \right)_{ij} + (M_F)_{ij} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{1}{2} \left(\left((M_F)^T \, (y_1)^\dagger \, y_1 \right)_{ij} + \left((y_1)^T \, (y_1)^* \, M_F \right)_{ij} + \left((M_F)^T \, (y_2)^\dagger \, y_2 \right)_{ij} + \left((y_2)^T \, (y_2)^* \, M_F \right)_{ij} \right)$	$- 6 g_2^2 \left(\left((M_F)^T \right)_{ij} + (M_F)_{ij} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: M_F

Terms 3 of 4

Column	Representative term structure
T1	$\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^{*} M_F\right)_{i j}+3\left(M_F\left(y_2\right)^{\dagger} y_2\right)_{i j})$
T2	$-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$
T3	$-\frac{3}{2} g_1^2\left(M_F\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^{*} M_F\right)_{i j}+3\left(M_F\left(y_2\right)^{\dagger} y_2\right)_{i j})-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$		$-\frac{3}{2} g_1^2\left(M_F\right)_{i j}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^{*} M_F\right)_{i j}+2\left(M_F\left(y_2\right)^{\dagger} y_2\right)_{i j})-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$		$-\frac{3}{2} g_1^2\left(M_F\right)_{i j}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2} g_2^2\left(M_F\right)_{i j}$	$-\frac{3}{2} g_1^2\left(M_F\right)_{i j}$

Quantity: M_F

Terms 4 of 4

Column	Representative term structure
T1	$\frac{3}{4} \left(\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij} + \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3}{4} \left(\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij} + \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij} \right)$

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\,\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$
T2	$-8\,\mathrm{Tr}\Big(M_F\,(M_F)^*\,(y_1)^T\,(y_1)^*\Big)$
T3	$-8\,\mathrm{Tr}\Big(M_F\,(y_1)^\dagger\,y_1\,(M_F)^*\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-8\,\mathrm{Tr}\Big(M_F\,(M_F)^*\,(y_1)^T\,(y_1)^*\Big)$	$-8\,\mathrm{Tr}\Big(M_F\,(y_1)^\dagger\,y_1\,(M_F)^*\Big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-12\,\mathrm{Tr}\Big(M_F\,(M_F)^*\,(y_1)^T\,(y_1)^*\Big)$	$-12\,\mathrm{Tr}\Big(M_F\,(y_1)^\dagger\,y_1\,(M_F)^*\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: m_1^2

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9g_2^2m_1^2}{2}$
T2	$-\frac{3g_1^2m_1^2}{2}$
T3	$+4\lambda_{12}^{(1)}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	$+4\lambda_{12}^{(1)}m_2^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_1^2}{2}$	$-\frac{3g_1^2m_1^2}{2}$	$+4\lambda_{12}^{(1)}m_2^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+6\lambda_{12}^{(1)}m_2^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)}m_2^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: m_1^2

Terms 3 of 5

Column	Representative term structure
T1	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$
T2	$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$
T3	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$

Quantity: m_1^2

Terms 4 of 5

Column	Representative term structure
T1	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$
T2	$-12g_2^2m_1^2$
T3	$-6g_1^2m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$	$-6g_1^2m_1^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$	—

Quantity: m_1^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T2	$-8\,\text{Tr}\Big(M_F\,(M_F)^*\,(y_2)^T\,(y_2)^*\Big)$
T3	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^*\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^*\,(y_2)^T\,(y_2)^*\Big)$	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^*\Big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-12\,\text{Tr}\Big(M_F\,(M_F)^*\,(y_2)^T\,(y_2)^*\Big)$	$-12\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^*\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: m_2^2

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{9g_2^2m_2^2}{2}$
T2	$-\frac{3g_1^2m_2^2}{2}$
T3	$+4\lambda_{12}^{(1)}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	$+4\lambda_{12}^{(1)}m_1^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_2^2}{2}$	$-\frac{3g_1^2m_2^2}{2}$	$+4\lambda_{12}^{(1)}m_1^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)}m_1^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+6\lambda_{12}^{(1)}m_1^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: m_2^2

Terms 3 of 5

Column	Representative term structure
T1	$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$
T2	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$
T3	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-8\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4\text{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)$

Quantity: m_2^2

Terms 4 of 5

Column	Representative term structure
T1	$-12g_2^2m_2^2$
T2	$-6g_1^2m_2^2$
T3	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	—

Quantity: m_2^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$$\beta_{\lambda_1}^{(1)}$$

Quantity: λ_1

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$
T3	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Quantity: λ_1

Terms 2 of 7

Column	Representative term structure
T1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$
T2	$+8 \lambda_7 \left(\lambda_7\right)^{*}$
T3	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+8 \lambda_7 \left(\lambda_7\right)^{*}$	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}\right)^{*}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+8 \lambda_7 \left(\lambda_7\right)^{*}$	$+4 \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}\right)^{*}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	—	—

Terms 3 of 7

Column	Representative term structure
T1	$+8\lambda_8\left(\lambda_8\right)^*$
T2	$+4\lambda_5\left(\lambda_5\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_8\left(\lambda_8\right)^*$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_8\left(\lambda_8\right)^*$	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$

Terms 4 of 7

Column	Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$
T2	$-9g_2^2\lambda_1$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Quantity: λ_1

Terms 5 of 7

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+4\lambda_{HS_1}^{(1)2}$
T3	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	—

Quantity: λ_1

Terms 6 of 7

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$
T3	$+3\lambda_{HS_2}^{(A) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{HS_2}^{(1) \ 2}$	—	$+3\lambda_{HS_2}^{(A) \ 2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{HS_2}^{(1) \ 2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{HS_2}^{(1) \ 2}$	—	$+3\lambda_{HS_2}^{(A) \ 2}$

Quantity: λ_1

Terms 7 of 7

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Terms 1 of 9

Column	Representative term structure
T1	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$
T2	$+ \lambda_{11}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\left(\lambda_{11}\right)^*)$
T3	$+2\left(\lambda_{11}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$+ \lambda_{11}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\left(\lambda_{11}\right)^*)$	$+2\left(\lambda_{11}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$+ \lambda_{11}(3\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\left(\lambda_{11}\right)^*)$	$+3\left(\lambda_{11}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T2	$+8\lambda_7\left(\lambda_7\right)^*$
T3	$+8\lambda_9\left(\lambda_9\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 9

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)} \right)$
T3	$-9g_2^2 \lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)} \right)$	$-9g_2^2 \lambda_{S_1}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)} \right)$	$-9g_2^2 \lambda_{S_1}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+\frac{3g_1^4}{4}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_{12}^{(1)2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)2}$
T2	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$
T3	$+12\lambda_{S_1}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_1}^{(1)2}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	$+12\lambda_{S_1}^{(1)2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_1}^{(1)2}$	$+\frac{\lambda_{HS_1}^{(A)2}}{4}$	$+12\lambda_{S_1}^{(1)2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)2}$	—	$+10\lambda_{S_1}^{(1)2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)2}$	—	$+14\lambda_{S_1}^{(1)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)2}$	—	$+14\lambda_{S_1}^{(1)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$
T2	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$
T3	$+21g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 7 of 9

Column	Representative term structure	
T1	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	
T2	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	
T3	$+12g_1^4$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+12g_1^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$
T2	$+26\lambda_{S_1}^{(A)^2}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$	$+26\lambda_{S_1}^{(A)^2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$	$+26\lambda_{S_1}^{(A)^2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+6\lambda_{12}^{(A)^2}$
T2	$+6\lambda_{12}^{(\times)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(A)^2}$	$+6\lambda_{12}^{(\times)^2}$

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}^{(1)}}^{(1)}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{10}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\ (\lambda_{10})^*)$
T2	$+2\ (\lambda_{10})^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3	$+4\lambda_{S_2}^{(1)}T_\nu^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{10}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\ (\lambda_{10})^*)$	$+2\ (\lambda_{10})^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$	$+4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{10}(3\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\ (\lambda_{10})^*)$	$+3\ (\lambda_{10})^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$	$+6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T2	$+8\lambda_8\left(\lambda_8\right)^*$
T3	$+8\lambda_9\left(\lambda_9\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-5\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$+8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-8\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 9

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)} \right)$
T3	$-9g_2^2 \lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)} \right)$	$-9g_2^2 \lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)} \right)$	$-9g_2^2 \lambda_{S_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+\frac{3g_1^4}{4}$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_1^4}{4}$	$+4\lambda_{12}^{(1)2}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_{12}^{(1)2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$
T3	$+12\lambda_{S_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$+12\lambda_{S_2}^{(1) \ 2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$+12\lambda_{S_2}^{(1) \ 2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+14\lambda_{S_2}^{(1) \ 2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+10\lambda_{S_2}^{(1) \ 2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+14\lambda_{S_2}^{(1) \ 2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$
T2	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$
T3	$+21g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$+21g_2^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$+21g_2^4$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 7 of 9

Column	Representative term structure
T1	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$
T2	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3	$+12g_1^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+12g_1^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+12g_1^4$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$
T2	$+26\lambda_{S_2}^{(A)^2}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+26\lambda_{S_2}^{(A)^2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+26\lambda_{S_2}^{(A)^2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+6\lambda_{12}^{(A)^2}$
T2	$+6\lambda_{12}^{(\times)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(A)^2}$	$+6\lambda_{12}^{(\times)^2}$

$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)\big)$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$
T3	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big)\big)$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$	$+ \frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T2	$+24\lambda_7\left(\lambda_7\right)^*$
T3	$+3\lambda_{11}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+24\lambda_7\left(\lambda_7\right)^*$	$+3\lambda_{11}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+24\lambda_7\left(\lambda_7\right)^*$	$+3\lambda_{11}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 9

Column	Representative term structure
T1	$+6\lambda_5\left(\lambda_5\right)^*$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_5\left(\lambda_5\right)^*$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_5\left(\lambda_5\right)^*$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_5\left(\lambda_5\right)^*$	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+ \frac{3g_1^4}{4}$
T2	$-3g_1^2 \lambda_{HS_1}^{(1)}$
T3	$+4\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_1^4}{4}$	$-3g_1^2 \lambda_{HS_1}^{(1)}$	$+4\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_1^4}{4}$	$-3g_1^2 \lambda_{HS_1}^{(1)}$	$+4\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+6\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$+6\lambda_1\lambda_{HS_1}^{(1)}$
T2	$+4\lambda_{HS_1}^{(1)^2}$
T3	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$
T2	$\lambda_{HS_1}^{(1)}(3T_\nu^{(1)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T3	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{4}$	$\lambda_{HS_1}^{(1)}(3T_\nu^{(1)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 7 of 9

Column	Representative term structure
T1	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T2	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{(A)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$+6g_2^4$
T2	$+3g_1^4$
T3	$+6\lambda_{HS_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6g_2^4$	$+3g_1^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	—	$+6\lambda_{HS_1}^{(A)2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+ \lambda_{HS_2}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{HS_2}^{(1)} (6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)})$

$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)\big)$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)$
T3	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)\big)$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)$	$+ \frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)$	$+ \frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$
T2	$+3\lambda_{10}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^{*}$
T3	$+24\lambda_8\left(\lambda_8\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+3\lambda_{10}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^{*}$	$+24\lambda_8\left(\lambda_8\right)^{*}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+3\lambda_{10}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^{*}$	$+24\lambda_8\left(\lambda_8\right)^{*}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-4\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 9

Column	Representative term structure
T1	$+6\lambda_5\left(\lambda_5\right)^*$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_5\left(\lambda_5\right)^*$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_5\left(\lambda_5\right)^*$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_5\left(\lambda_5\right)^*$	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+ \frac{3g_1^4}{4}$
T2	$-3g_1^2 \lambda_{HS_2}^{(1)}$
T3	$+4\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_1^4}{4}$	$-3g_1^2 \lambda_{HS_2}^{(1)}$	$+4\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3g_1^4}{4}$	$-3g_1^2 \lambda_{HS_2}^{(1)}$	$+4\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+6\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T2	$+4\lambda_{HS_2}^{(1)^2}$
T3	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$
T2	$\lambda_{HS_2}^{(1)}(3T_\nu^{(2)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T3	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{4}$	$\lambda_{HS_2}^{(1)}(3T_\nu^{(2)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{S_2}^{(A)})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 7 of 9

Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$
T3	$+3g_1^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T2	$+6\lambda_{HS_2}^{(A)2}$
T3	$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{HS_2}^{(A)2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$	—	$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$	$+6\lambda_{HS_2}^{(A)2}$	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$16\pi^2\,\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$

Terms 1 of 11

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$
T2	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\Big(y_1\ (y_2)^{\dagger}\Big)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$
T3	$+\frac{3}{2}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\Big(y_2\ (y_1)^{\dagger}\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\Big(y_1\ (y_2)^{\dagger}\Big)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$ $+\lambda_{11}\left(\frac{3}{2}\text{Tr}\Big(y_1\ (y_2)^{\dagger}\Big)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$	$+\frac{3}{2}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\Big(y_2\ (y_1)^{\dagger}\Big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{10}\left(\frac{9}{4}\text{Tr}\Big(y_1\ (y_2)^{\dagger}\Big)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$ $+\lambda_{11}\left(\frac{9}{4}\text{Tr}\Big(y_1\ (y_2)^{\dagger}\Big)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$	$+\frac{9}{4}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\Big(y_2\ (y_1)^{\dagger}\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 11

Column	Representative term structure
T1	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T2	$-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T3	$+2\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^{*}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+2\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^{*}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-3\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+2\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\right)^{*}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 11

Column	Representative term structure
T1	$+24\lambda_9\left(\lambda_9\right)^*$
T2	$+3\lambda_5\left(\lambda_5\right)^*$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_9\left(\lambda_9\right)^*$	$+3\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_9\left(\lambda_9\right)^*$	$+3\lambda_5\left(\lambda_5\right)^*$	$+\frac{9g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_5\left(\lambda_5\right)^*$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_5\left(\lambda_5\right)^*$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 11

Column	Representative term structure
T1	$-9g_2^2\lambda_{12}^{(1)}$
T2	$+\frac{3g_1^4}{4}$
T3	$-3g_1^2\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{12}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{12}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{12}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	—	$-6g_1^2\lambda_{12}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	—	$-6g_1^2\lambda_{12}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 11

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)2}$
T2	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^{(1)2}$	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 11

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3	$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3\lambda_{12}^{(A)2}}{4}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3\lambda_{12}^{(A)2}}{4}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 7 of 11

Column	Representative term structure
T1	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T2	$\lambda_{12}^{(1)}\left(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)}\right)$
T3	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{12}^{(1)}\left(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)}\right)$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 8 of 11

Column	Representative term structure
T1	$\lambda_{12}^{(1)} \left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)} \right)$
T2	$+ \lambda_{12}^{(\times)} \left(\frac{4}{3} T_{\nu}^{(1)} + \frac{4}{3} T_{\nu}^{(2)} \right)$
T3	$+ \frac{4}{3} \lambda_5 \left(\lambda_5 \right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}^{(1)} \left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)} \right) + \lambda_{12}^{(\times)} \left(\frac{4}{3} T_{\nu}^{(1)} + \frac{4}{3} T_{\nu}^{(2)} \right)$		$+ \frac{4}{3} \lambda_5 \left(\lambda_5 \right)^*$

Quantity: $\lambda_{12}^{(1)}$

Terms 9 of 11

Column	Representative term structure	
T1	$+16g_2^4$	
T2	$+g_2^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$	
T3	$+g_1^2(-6\lambda_{12}^{(1)} - 4\lambda_{12}^{(\times)})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$	$+g_1^2(-6\lambda_{12}^{(1)} - 4\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(1)}$

Terms 10 of 11

Column	Representative term structure
T1	$+ \lambda_{S_1}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$
T2	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$
T3	$+ 16 \lambda_{12}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{S_1}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ 16 \lambda_{12}^{(A)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 11 of 11

Column	Representative term structure
T1	$+\frac{32\lambda_{12}^{(\times)2}}{3}$
T2	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$
T3	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$
T2	$+2\left(\lambda_7\right)^*\mathrm{Tr}\left(y_2\,\left(y_1\right)^\dagger\right)$
T3	$+2\left(\lambda_8\right)^*\mathrm{Tr}\left(y_1\,\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_3\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$	$+2\left(\lambda_7\right)^*\mathrm{Tr}\left(y_2\,\left(y_1\right)^\dagger\right)$	$+2\left(\lambda_8\right)^*\mathrm{Tr}\left(y_1\,\left(y_2\right)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+3\left(\lambda_7\right)^*\mathrm{Tr}\left(y_2\,\left(y_1\right)^\dagger\right)$	$+3\left(\lambda_8\right)^*\mathrm{Tr}\left(y_1\,\left(y_2\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T1	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_7)^*$
T2	$+4\lambda_{11} (\lambda_7)^*$
T3	$+4 (\lambda_8)^* ((\lambda_{10})^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_7)^*$	$+4\lambda_{11} (\lambda_7)^*$	$+4 (\lambda_8)^* ((\lambda_{10})^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_7)^*$	$+4\lambda_{11} (\lambda_7)^*$	$+4 (\lambda_8)^* ((\lambda_{10})^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_5

Terms 3 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_5$
T2	$-3g_1^2\lambda_5$
T3	$+2\lambda_{12}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_5$	$-3g_1^2\lambda_5$	$+2\lambda_{12}^{(1)}\lambda_5$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_5$	$-3g_1^2\lambda_5$	$+2\lambda_{12}^{(1)}\lambda_5$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)}\lambda_5$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+2\lambda_{12}^{(1)}\lambda_5$

Quantity: λ_5

Terms 4 of 6

Column	Representative term structure
T1	$+2\lambda_1\lambda_5$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$	$+4\lambda_{HS_2}^{(1)}\lambda_5$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$	$+4\lambda_{HS_2}^{(1)}\lambda_5$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$	$+4\lambda_{HS_2}^{(1)}\lambda_5$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$	$+4\lambda_{HS_2}^{(1)}\lambda_5$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Quantity: λ_5

Terms 5 of 6

Column	Representative term structure
T1	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2 \lambda_{HS1}^{(A)}}{3} + 2 \lambda_{HS2}^{(A)} \right)$
T2	$\lambda_5 (2 T_\nu^{(1)} + T_\nu^{(2)} + 6 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) + 4 \lambda_{HS2}^{(A)})$
T3	$-\frac{21 g_2^2 \lambda_5}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2 \lambda_{HS1}^{(A)}}{3} + 2 \lambda_{HS2}^{(A)} \right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$\lambda_5 (2 T_\nu^{(1)} + T_\nu^{(2)} + 6 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) + 4 \lambda_{HS2}^{(A)})$	$-\frac{21 g_2^2 \lambda_5}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-\frac{21 g_2^2 \lambda_5}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{33 g_2^2 \lambda_5}{2}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{9g_1^2\lambda_5}{2}$
T2	$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS_1}^{(A)})$
T3	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_1^2\lambda_5}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{9g_1^2\lambda_5}{2}$	$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-4\lambda_{HS_1}^{(A)})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_1^2\lambda_5}{2}$	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3}\right)$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$
T2	$+ \lambda_{10}\left(\frac{2}{3}\text{Tr}\left(y_1\ (y_2)^{\dagger}\right)+\frac{4}{3}(5\left(\lambda_{10}\right)^*+(\lambda_{11})^*)\right)$
T3	$+ \lambda_{11}\left(\frac{2}{3}\text{Tr}\left(y_1\ (y_2)^{\dagger}\right)+\frac{4}{3}((\lambda_{10})^*+5\left(\lambda_{11}\right)^*)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$+ \lambda_{10}\left(\frac{2}{3}\text{Tr}\left(y_1\ (y_2)^{\dagger}\right)+\frac{4}{3}(5\left(\lambda_{10}\right)^*+(\lambda_{11})^*)\right)$	$+ \lambda_{11}\left(\frac{2}{3}\text{Tr}\left(y_1\ (y_2)^{\dagger}\right)+\frac{4}{3}((\lambda_{10})^*+5\left(\lambda_{11}\right)^*)\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 8

Column	Representative term structure
T1	$+\frac{2}{3}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T2	$-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$
T3	$+\frac{64}{3}\lambda_9\ (\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2}{3}((\lambda_{10})^*+(\lambda_{11})^*)\text{Tr}\big(y_2\ (y_1)^\dagger\big)$	$-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{64}{3}\lambda_9\ (\lambda_9)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	$+\frac{64}{3}\lambda_9\ (\lambda_9)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$	—

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 8

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$
T3	$-9g_2^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	$-9g_2^2\lambda_{12}^{(A)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$	—	$-24g_2^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 8

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T3	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 5 of 8

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T2	$\lambda_{12}^{(A)}(3T_\nu^{(1)}+3T_\nu^{(2)})$
T3	$+\lambda_{10}\left(\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)+\frac{4}{3}(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	$\lambda_{12}^{(A)}(3T_\nu^{(1)}+3T_\nu^{(2)})$	$+\lambda_{10}\left(\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)+\frac{4}{3}(5\left(\lambda_{10}\right)^*+\left(\lambda_{11}\right)^*)\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 6 of 8

Column	Representative term structure
T1	$+ \lambda_{11} \left(\text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \left(\lambda_{11} \right)^*) \right)$
T2	$+ ((\lambda_{10})^* + (\lambda_{11})^*) \text{Tr} \left(y_2 \left(y_1 \right)^{\dagger} \right)$
T3	$- \frac{8}{3} \text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} y_2 \left(y_1 \right)^{\dagger} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{11} \left(\text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \left(\lambda_{11} \right)^*) \right)$	$+ ((\lambda_{10})^* + (\lambda_{11})^*) \text{Tr} \left(y_2 \left(y_1 \right)^{\dagger} \right)$	$- \frac{8}{3} \text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} y_2 \left(y_1 \right)^{\dagger} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 7 of 8

Column	Representative term structure
T1	$\lambda_{12}^{(A)} \left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} \right)$
T2	$-6g_1^2 \lambda_{12}^{(A)}$
T3	$+16\lambda_{12}^{(A)} \lambda_{S_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}^{(A)} \left(2T_{\nu}^{(1)} + 2T_{\nu}^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} \right)$	$-6g_1^2 \lambda_{12}^{(A)}$	$+16\lambda_{12}^{(A)} \lambda_{S_1}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 8 of 8

Column	Representative term structure
T1	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$
T2	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta_{\lambda_{12}^{(\times)}}^{(1)}$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(\times)}}{d\mu}=\beta_{\lambda_{12}^{(\times)}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$
T2	$+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$
T3	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+\frac{1}{2}T_\nu^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)+\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$		$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 5

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 5

Column	Representative term structure
T1	$+g_1^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$
T2	$+\lambda_{12}^{(1)2}$
T3	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T2	$+4\lambda_{12}^{(A)2}$
T3	$+11\lambda_{12}^{(\times)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 5

Column	Representative term structure
T1	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T2	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T3	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\,\text{Tr}\big(y_1\,(y_2)^\dagger\big)-\frac{4}{3}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)\right)$
T3	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\,\text{Tr}\big(y_1\,(y_2)^\dagger\big)-\frac{4}{3}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)\right)$	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$
T2	$+\frac{64}{3}\lambda_7\ (\lambda_7)^*$
T3	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$	$+\frac{64}{3}\lambda_7\ (\lambda_7)^*$	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{4}{3}\text{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$	$+\frac{64}{3}\lambda_7\ (\lambda_7)^*$	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{4}{3}\text{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{4}{3}\text{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3	$-9g_2^2\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$	$-9g_2^2\lambda_{HS_1}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$	$-9g_2^2\lambda_{HS_1}^{(A)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T3	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T2	$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*)\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T2	$\lambda_{HS_1}^{(A)}\left(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{(A)}\right)$
T3	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\lambda_{HS_1}^{(A)}\left(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{(A)}\right)+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{HS_1}^{(A)}\left(2T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{(A)}\right)$	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T2	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)-4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*))$
T3	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)-4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*))$	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-3\operatorname{Tr}\big(y_1\ (y_2)^\dagger\big)-4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{10})^*))$	$-3\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\operatorname{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+8\,\mathrm{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T2	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\right)^*$
T3	$+64\lambda_8\,(\lambda_8)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\,\mathrm{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\right)^*$	$+64\lambda_8\,(\lambda_8)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-4\,\mathrm{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\right)^*$	$+64\lambda_8\,(\lambda_8)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+16\lambda_5\left(\lambda_5\right)^*$
T2	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3	$-9g_2^2\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+16\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+16\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T2	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
T2	$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 6 of 7

Column	Representative term structure	
T1	$+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	
T2	$+\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	
T3	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$+\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$
T2	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

$\beta_{\lambda_7}^{(1)}$
Quantity: λ_7

$$16\pi^2\,\mu\,\frac{d\lambda_7}{d\mu}=\beta_{\lambda_7}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_7(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+(\lambda_5)^*\text{Tr}\big(y_2\,(y_1)^\dagger\big)$
T3	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_7(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+(\lambda_5)^*\text{Tr}\big(y_2\,(y_1)^\dagger\big)$	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_7

Terms 2 of 5

Column	Representative term structure
T1	$+2\lambda_{11}(\lambda_5)^*$
T2	$-9g_2^2\lambda_7$
T3	$-3g_1^2\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{11}(\lambda_5)^*$	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{11}(\lambda_5)^*$	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_7

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda_1\lambda_7$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_7$
T3	$+\frac{4\lambda_{HS_1}^{(A)}}{3}\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_7$	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}}{3}\lambda_7$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_1\lambda_7$	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}}{3}\lambda_7$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_7

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_7\lambda_{S_1}^{(1)}$
T2	$+4\lambda_8\lambda_9$
T3	$\lambda_7(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$	$\lambda_7(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_7

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{\lambda_{12}}^{(1)}$

Quantity: λ_{12}

$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}\left(\frac{1}{2}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$
T2	$+\lambda_{HS_1}^{(A)}\left(-\frac{1}{8}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$
T3	$+\lambda_{HS_2}^{(1)}\left(\frac{1}{2}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_1}^{(1)}\left(\frac{1}{2}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$	$+\lambda_{HS_1}^{(A)}\left(-\frac{1}{8}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$	$+\lambda_{HS_2}^{(1)}\left(\frac{1}{2}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HS_1}^{(1)}\left(\frac{3}{4}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$	$+\lambda_{HS_1}^{(A)}\left(-\frac{3}{16}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$	$+\lambda_{HS_2}^{(1)}\left(\frac{3}{4}\,\text{Tr}\big(y_2\,(y_1)^\dagger\big)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(A)}\left(-\frac{3}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$
T3	$-4\text{Tr}\left(y_2\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HS_2}^{(A)}\left(-\frac{3}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$	$-4\text{Tr}\left(y_2\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HS_2}^{(A)}\left(-\frac{9}{16}\text{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{3\lambda_{10}}{4}\right)$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T3	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_9\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$-6\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{12}

Terms 4 of 5

Column	Representative term structure
T1	$-3g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_1\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$-6\lambda_7\lambda_5$
T2	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-6\lambda_7\lambda_5$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-6\lambda_7\lambda_5$	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) + 2 \text{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) + 6 \text{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_8}^{(1)}$

Quantity: λ_8

$$16\pi^2\,\mu\,\frac{d\lambda_8}{d\mu}=\beta_{\lambda_8}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_8(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+(\lambda_5)^*\text{Tr}\big(y_1\,(y_2)^\dagger\big)$
T3	$+4\lambda_7\,(\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_8(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+(\lambda_5)^*\text{Tr}\big(y_1\,(y_2)^\dagger\big)$	$+4\lambda_7\,(\lambda_9)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+4\lambda_7\,(\lambda_9)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_8

Terms 2 of 4

Column	Representative term structure	
T1	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-{\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)}^*\right)$	
T2	$-9g_2^2\lambda_8$	
T3	$-3g_1^2\lambda_8$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-{\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)}^*\right)$	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-{\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)}^*\right)$	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_8

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_1\lambda_8$
T2	$+8\lambda_{HS_2}^{(1)}\lambda_8$
T3	$+4\lambda_{HS_2}^{(A)}\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_8$	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_1\lambda_8$	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_8

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_8\lambda_{S_2}^{(1)}$
T2	$\lambda_8(3T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3	$+\frac{3}{2}\left(\lambda_5\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_8\lambda_{S_2}^{(1)}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_8\lambda_{S_2}^{(1)}$	$\lambda_8(3T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+\frac{3}{2}\left(\lambda_5\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	
T2	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	
T3	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 6

Column	Representative term structure
T1	$+ \frac{3g_2^4}{4}$
T2	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$
T3	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 6

Column	Representative term structure
T1	$+\frac{3g_1^4}{2}$
T2	$+\frac{\lambda_{12}^{(1)2}}{4}$
T3	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3g_1^4}{2}$	$+\frac{\lambda_{12}^{(1)2}}{4}$	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$
T2	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$
T3	$+22\lambda_{S_1}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$	$+\frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 5 of 6

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^{(1)2}}{4}$
T2	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3	$+12\lambda_{12}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{\lambda_{12}^{(\times)^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{(\times)^2}}{2}$

$\beta_{\lambda_{11}}^{(1)}$

Quantity: λ_{11}

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu}=\beta_{\lambda_{11}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$
T3	$+\lambda_{S_1}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+12\lambda_{11})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+\lambda_{S_1}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+12\lambda_{11})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{12}^{(A)}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{11}

Terms 2 of 5

Column	Representative term structure
T1	$+ \lambda_9 (2 \operatorname{Tr} \left(y_1 \ (y_2)^{\dagger} \right) + 4 ((\lambda_{10})^* + 5 \ (\lambda_{11})^*))$
T2	$+ \lambda_{11} (3 T_{\nu}^{(1)} + T_{\nu}^{(2)})$
T3	$- 4 \operatorname{Tr} \left(y_1 \ (y_1)^{\dagger} \ y_2 \ (y_1)^{\dagger} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_9 (2 \operatorname{Tr} \left(y_1 \ (y_2)^{\dagger} \right) + 4 ((\lambda_{10})^* + 5 \ (\lambda_{11})^*))$	$+ \lambda_{11} (3 T_{\nu}^{(1)} + T_{\nu}^{(2)})$	$- 4 \operatorname{Tr} \left(y_1 \ (y_1)^{\dagger} \ y_2 \ (y_1)^{\dagger} \right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_9 (3 \operatorname{Tr} \left(y_1 \ (y_2)^{\dagger} \right) + 4 ((\lambda_{10})^* + 5 \ (\lambda_{11})^*))$	—	$- 5 \operatorname{Tr} \left(y_1 \ (y_1)^{\dagger} \ y_2 \ (y_1)^{\dagger} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{11}

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_{11}$
T2	$-3g_1^2\lambda_{11}$
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{11}

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_7\lambda_5$
T3	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{11}

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_1}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{11} \right)$
T2	$+ \lambda_{11} \left(\frac{9}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{11} \right)$	$+ \lambda_{11} \left(\frac{9}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_9}^{(1)}$
Quantity: λ_9

$$16\pi^2\,\mu\,\frac{d\lambda_9}{d\mu}=\beta_{\lambda_9}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$
T2	$+ \lambda_{10}(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+2\lambda_{11})$
T3	$+ \lambda_{11}\text{Tr}\left(y_2\ (y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	$+ \lambda_{10}(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+2\lambda_{11})$	$+ \lambda_{11}\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_9

Terms 2 of 6

Column	Representative term structure
T1	$-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T2	$+4\lambda_7\left(\lambda_8\right)^*$
T3	$-9g_2^2\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_9

Terms 3 of 6

Column	Representative term structure
T1	$-3g_1^2\lambda_9$
T2	$+8\lambda_{12}^{(1)}\lambda_9$
T3	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_1^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_1^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_9

Terms 4 of 6

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$
T2	$+2\lambda_{S_1}^{(1)}\lambda_9$
T3	$+2\lambda_9\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_9

Terms 5 of 6

Column	Representative term structure
T1	$+5\lambda_{10}^2$
T2	$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$
T3	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{11}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$	$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{11}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_9

Terms 6 of 6

Column	Representative term structure
T1	$+\frac{3}{2}\lambda_{11}\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
T2	$-\frac{5}{2}\text{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\lambda_{11}\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$	$-\frac{5}{2}\text{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{10}}^{(1)}$

Quantity: λ_{10}

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu}=\beta_{\lambda_{10}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}^{(1)}\left(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T2	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$
T3	$+\lambda_9(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(1)}\left(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	$+\lambda_9(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{12}^{(A)}\left(\frac{1}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	$+\lambda_9(3\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{10}

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}^{(1)}(\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+12\lambda_{10})$
T2	$+\lambda_{10}\left(T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$
T3	$-4\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{S_2}^{(1)}(\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+12\lambda_{10})$	$+\lambda_{10}\left(T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$	$-4\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-5\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{10}

Terms 3 of 5

Column	Representative term structure
T1	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T2	$-9g_2^2\lambda_{10}$
T3	$-3g_1^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{10}$	$-3g_1^2\lambda_{10}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{10}$	$-3g_1^2\lambda_{10}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Quantity: λ_{10}

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_2}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{10} \right)$
T2	$+ \lambda_{10} \left(\frac{3}{2} T_\nu^{(1)} + \frac{9}{2} T_\nu^{(2)} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_2}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{10} \right)$	$+ \lambda_{10} \left(\frac{3}{2} T_\nu^{(1)} + \frac{9}{2} T_\nu^{(2)} \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—

$\beta_{\lambda_{S_2}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{(A)}}{d\mu}=\beta_{\lambda_{S_2}^{(A)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$
T2	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$
T3	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 6

Column	Representative term structure
T1	$+ \frac{3g_2^4}{4}$
T2	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$
T3	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_2}^{(1)}}{2} - 12\lambda_{S_2}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_2}^{(1)} - 24\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 6

Column	Representative term structure
T1	$+ \frac{3g_1^4}{2}$
T2	$+ \frac{\lambda_{12}^{(1)2}}{4}$
T3	$+ \frac{\lambda_{HS_2}^{(1)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_1^4}{2}$	$+ \frac{\lambda_{12}^{(1)2}}{4}$	$+ \frac{\lambda_{HS_2}^{(1)2}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_1^4}{2}$	—	$+ \frac{\lambda_{HS_2}^{(1)2}}{2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{HS_2}^{(A) \ 2}$
T2	$+\frac{7\lambda_{S_2}^{(1)2}}{4}$
T3	$+22\lambda_{S_2}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1)2}}{4}$	$+22\lambda_{S_2}^{(A)2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1)2}}{4}$	$+22\lambda_{S_2}^{(A)2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 6

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^{(1)2}}{4}$
T2	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3	$+12\lambda_{12}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{\lambda_{12}^{(\times)^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{(\times)^2}}{2}$